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Question 1

- a) Let $S = \text{SPayGlobal}$
 Let $G = \text{GrabPay}$
 Let $T = \text{TNG eWallet}$

$$|S| = 340$$

$$|G| = 180$$

$$|T| = 190$$

$$|S \cap G| = 110$$

$$|S \cap T| = 60$$

$$|S \cap G \cap T| = 40$$

$$|S^c \cap G^c \cap T^c| = 20$$

$$|G \cap T| = x$$

$$|S \cup G \cup T| = |S| + |G| + |T| - |S \cap G| - |S \cap T| - |G \cap T| + |S \cap G \cap T|$$

$$(500 - 20) = 340 + 180 + 190 - 110 - 60 - x + 40$$

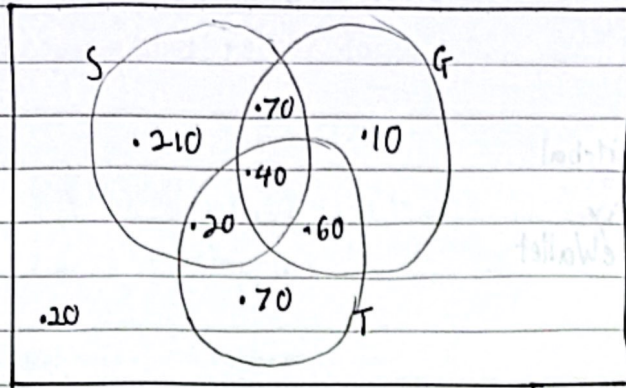
$$480 = 580 - x$$

$$x = 580 - 480$$

$$x = 100$$

$$|G \cap T| = 100$$

b)



Students who only use

$$S \text{ and } G = 110 - 40 \\ = 70$$

$$S \text{ and } T = 60 - 40 \\ = 20$$

$$G \text{ and } T = 100 - 40 \\ = 60$$

$$S = 340 - 70 - 40 - 20 \\ = 210$$

$$G = 180 - 70 - 40 - 60 \\ = 10$$

$$T = 190 - 20 - 40 - 60 \\ = 70$$

$$c) |T^c| = U - |T|$$

$$= 500 - 70 - 20 - 40 - 60 \\ = 310$$

$$ii) |S \cap G^c| = 210 + 20 \\ = 230$$

$$iii) \text{Students who only use } S + \text{only use } G + \text{only use } T \\ = 210 + 10 + 70 \\ = 290$$

$$di) (T^c \cap G^c)$$

$$ii) (S \cap G) \cap T'$$

2ai) Vina enjoys shopping and neither Tasha enjoys watching K-dramas nor play video games.

ii) If Daniel plays video games, then either Tasha enjoys K-dramas or Vina does not enjoy shopping.

~~2aii) It is not the case that Vina enjoys shopping and Daniel plays video games.~~

iii) Vina does not enjoy shopping and Daniel doesn't play video games at the same time.

b) $g \rightarrow (k \wedge \neg s)$

ii) $\neg(k \vee g)$

iii) $k \leftrightarrow (s \wedge g)$

3) Let O = Programmers working in the office
 R = Programmers working remotely
 P = Programmers using office printer

O	R	P	$O \vee R$	$R \rightarrow \neg P$
T	T	T	T	F
T	T	F	T	T
T	F	T	T	T
T	F	F	T	T
F	T	T	T	F
F	T	F	T	T
F	F	T	F	T
F	F	F	F	T

Rows that satisfy all premises = true

$O = T$

$R = F$

$P = T$

~~$O = F$~~ Conclusion: $O = T$

Conclusion is true when all premises are true.

Hence, argument is valid.

4) R must satisfy the following to be an ~~equivalent~~ ^{equivalence} relation:

1. Reflexive
2. Symmetric
3. Transitive

1. Check if $(a, a) \in R$

(P, P)

(Q, Q)

(R, R)

(S, S)

(T, T)

(U, U)

Reflexive = Yes

2. Check if $(b, a) \in R$ for each $(a, b) \in R$

$(Q, R) \in R \rightarrow (R, Q) \in R$

$(S, T) \in R \rightarrow (T, S) \in R$

$(T, U) \in R \rightarrow (U, T) \in R$

All reverse so = symmetric

3. Check if $(a, b), (b, c) \in R$ then if $(a, c) \in R$

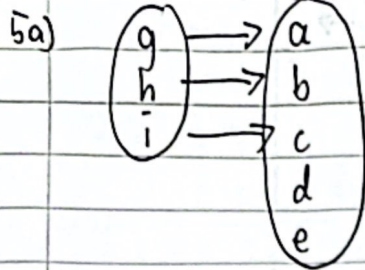
$(Q, R), (R, Q) \rightarrow (Q, Q) \in R$

$(R, Q), (Q, R) \rightarrow (R, R) \in R$

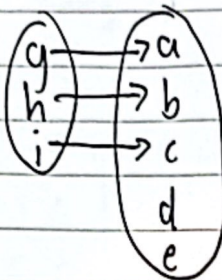
$(S, T), (T, U) \rightarrow (S, U) \notin R$

~~Not~~ Not transitive

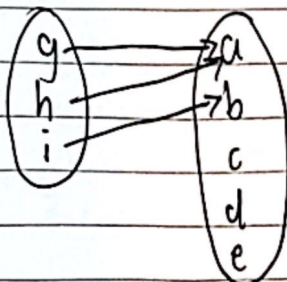
$\therefore R$ is not an equivalence relation



b) Injective



Not injective



Both g and h map to a. Hence, it is ~~not~~ not injective

c) There is not enough elements in the domain to cover all elements in ~~codomain~~ ^{codomain}, so there is no possibility of surjective. Hence, no bijective function exists.